

## ASTHMA: ROUTINE CARE

Ensure that a doctor confirms the diagnosis of asthma within 1 month. Refer the patient with newly diagnosed asthma for community health worker support, if available.

### Assess the patient with asthma

| Assess                               | When to assess                                                             | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Asthma symptoms to determine control | Every visit                                                                | <ul style="list-style-type: none"> <li>If wheeze, tight chest or difficulty breathing and no response to salbutamol inhaler, manage acute exacerbation ↗ 39.</li> <li>Any of the following indicate that the patient's asthma is <b>not controlled</b>: <ul style="list-style-type: none"> <li>Daytime cough, difficulty breathing or wheeze &gt; 2 times a week</li> <li>Night-time cough, wheeze, tight chest or difficulty breathing &gt; once a month</li> <li>Limitation of daily activities due to asthma symptoms</li> </ul> </li> <li>If none of above then asthma is <b>controlled</b>.</li> </ul> |
| Other symptoms                       | Every visit                                                                | <ul style="list-style-type: none"> <li>Manage symptoms as on symptom pages. Ask about and manage allergic rhinitis ↗ 34 and dyspepsia ↗ 44.</li> <li>If using inhaled corticosteroid and white patches on cheeks/gums/tongue/palate, <b>oral candida</b> likely ↗ 35.</li> </ul>                                                                                                                                                                                                                                                                                                                            |
| Adherence and inhaler technique      | Every visit                                                                | Check adherence and that patient is using inhaler and spacer correctly ↗ 123. If not adherent, give adherence support ↗ 173.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Peak expiratory flow rate (PEFR)     | At diagnosis, if symptoms worsening, if change to medication at last visit | Calculate % of predicted PEFR ↗ 124. If < 80%, asthma is <b>not controlled</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Health for All

↗ 116

### Advise the patient with asthma

- Advise to avoid triggers that may worsen asthma/hayfever (e.g. animals, cigarette smoke, dust, chemicals, pollen, grass), aspirin/NSAIDs (e.g. ibuprofen) and beta-blockers (e.g. atenolol).
- If patient smokes, encourage to stop ↗ 141. Support the patient to make a change ↗ 177.
- Ensure the patient understands medication: beta-agonist inhaler (salbutamol) relieves symptoms but does not control asthma. Inhaled corticosteroid (beclomethasone or fluticasone) prevents but does not relieve symptoms and it is the mainstay of treatment.
- Inhaled corticosteroids can cause oral candida: advise patient to rinse and gargle after each dose of inhaled corticosteroid.

### Treat the patient with asthma

- Give **influenza vaccine** 0.5mL IM yearly. Check that patient is up to date with COVID-19 vaccine.
- Give inhaled **salbutamol** 100-200mcg (1-2 puffs) 6-8 hourly, as needed. If exercise-related symptoms, advise patient to use **salbutamol** 200mcg (2 puffs) before exercise.
- If acute exacerbation was managed at this visit:
  - Give **prednisone** 40mg daily for a total of 7 days. If > 2 courses of oral prednisone given in past 6 months or exacerbation occurs on maximum treatment, also refer to doctor.
  - Only give antibiotic if fever or thick yellow/green sputum: give **amoxicillin**<sup>1</sup> 500mg 8 hourly for 5 days. If severe penicillin allergy<sup>1</sup>, give instead **azithromycin** 500mg daily for 3 days.
- Manage further according to asthma control:

#### Asthma not controlled or acute exacerbation

- Before stepping up treatment, ensure adherent and using inhaler/spacer correctly ↗ 123 and check patient is avoiding smoking, allergens and certain medications<sup>2</sup>.
- Give inhaled **beclomethasone** 200mcg 12 hourly. If already on it, increase dose to 400mcg 12 hourly.
- If still not controlled, doctor to stop beclomethasone and give instead inhaled **salmeterol/fluticasone**<sup>3</sup> 50/250mcg, 1 puff 12 hourly. If still not controlled after 3 months, refer.

#### Asthma controlled

- Continue inhaled medication at same dose.
- If controlled and no acute exacerbations for ≥ 6 months, step down treatment:
  - If on salmeterol/fluticasone, stop this and give instead **beclomethasone**<sup>3</sup> 400mcg 12 hourly.
  - If on beclomethasone, decrease dose to 200mcg 12 hourly. If already on 200mcg, stop beclomethasone.
  - If symptoms worsen, step up to same medication and dose when patient was controlled.

If asthma controlled, review 3 monthly. If not controlled, review monthly. If acute exacerbation, review after 1 week. **Advise to return before next appointment if symptoms persist/worsen.**

<sup>1</sup>History of anaphylaxis, urticaria or angioedema. <sup>2</sup>NSAIDs (aspirin/ibuprofen), beta blockers. <sup>3</sup>If on lopinavir/ritonavir or atazanavir/ritonavir, avoid **budesonide** and fluticasone, and discuss/refer instead.